

School of Computing and Informatics

Department of Computer Science

Course Title: Computer Graphics

Course Code: CoSc3072

Chapter one

Introduction to interactive computer graphics

Introduction

- Computer Graphics is a branch of Computer Science
 - Which has attracted some of the most **creative** people in the world
- The term Computer Graphics describes any use of computers to **create or manipulate images/pictures/animations** (or sequences of images)
 - This includes the **hardware** and **software** systems used to make these images

Cont....

- Computer graphics are visual representations of data shown on a monitor made on a computer.
- Computer graphics is an art of drawing **pictures, lines, charts**, etc. using computers with the help of programming.
- Computer graphics image is made up of number of **pixels**
- **Pixel** is the smallest addressable graphical unit represented on the computer screen.

Cont..

- The field of computer graphics has grown **extremely** over the past 10-20 years
 - Many **software systems** have been developed for generating computer graphics of various kinds
 - This can includes
 - Systems for **producing 3-dimensional** models of the scene to be drawn
 - **Rendering** software for drawing the images
 - The associated **user-interface** software and hardware.
- Today it is possible to produce images that are practically indistinguishable from photographic images

Types of Computer graphics

1. Interactive Computer Graphics
2. Non-Interactive Computer Graphics

Interactive Computer Graphics

- User Directly interact with graphics
- Interactive computer graphics affects our lives in a number of indirect ways
 - For example, it helps to train the pilots of our airplanes. We can create a **flight simulator** which **may help the pilots to get trained** not in a real aircraft, but on the grounds at the control of the flight simulator.

Non-Interactive Computer Graphics

- It is known as passive computer graphics
 - The computer graphics in which **user does not** have any kind of control over the image
- Image is only the product of **fixed** stored program and **will work according** to the instructions given in the program linearly.
 - The image is **totally under the control of program instructions** , not under the user.
 - Example: **screen savers**.

History Of Computer graphics

Reading Assignment

understand it by the following steps:

- **1940-1941:** The first directly digital computer-generated graphics that would associate with today as being actual CG. The very first radiosity image was invented at MIT in the 1940s.
- **1946:** The images were first presented at the 1946 national technical conference of the Illuminating Engineering Society of North America.
- **1948:** The images were published in the book: Lighting Design by Moon and D. E. Spencer. 1948.
- **1950:** John Whitney Sr. invents his computer-assisted mechanisms to create some of his graphic artwork and short films.
- **1951:** Vectorscope computer graphics display on the computer at MIT.
- The General Motors Research Laboratory also begins the study of computer-aided graphical design applications.

History

- **1955:** Sage system uses the first light pen as an input device at MIT Lab by **Bert Sutherland**.
- **1956:** Lawrence Livermore labs associate graphics display with IBM 704 and film recorder for color images.
- **Bertram Herzog** uses analog computers to create CRT graphics in the behavior of military vehicles at the University of Michigan computing center.
- **1957:** In the National Bureau of Standards first image-processed photo was used.
- The IBM 740 created a sequence of points on a CRT monitor to represent lines and shapes.
- **1958: Steven Coons, Ivan Sutherland, and Timothy Johnson** started working with the TX-2 computer system to manipulate the drawn pictures.**1959:** The first commercial film recorder produced in San Diego, CA.
- **Don Hart and Ed Jacks** invented the first computer-aided drawing system at General Motors Research Laboratory and IBM.

History

- **1960: William Fetter** was first termed "Computer Graphics" for cockpit drawing.
- **John Whitney Sr.** invents motion graphics in LA.
- **1962:** In MIT Lincoln Laboratory **Ivan Sutherland** produced a man-machine graphical communication system.
- **1963:** An analog computer was developed by **Charles Csuri** and used to transform a drawing.
- **Edgar Horwood** introduced a computer graphics mapping system which is used by U. S. Housing and urban development.
- **1965: IBM 2250**, the first graphics computer available.
- **1966: Ralph Baer** developed the first consumer computer graphics game, "Odyssey."
- **1968: Tony Pritchett** made the first computer animation "FLEXIPEDE" in the UK.
- **1972: Nolan Bushnell**, "the father of Electronic games," developed PONG game.

History

- **1973:** The concept of Z-buffer algorithm and texture mapping were developed by **Edwin Catmull**.
- **1974:** The Phong shading method is developed by **Phong Bui-Toung**.
- **1975:** Dr. Edwin Catmull introduced the Tween animation system.
- **1976:** The first 3D appearance film was created by **Gary Demos, John Whitey Jr. at Triple-I**.
- **1978:** For the mechanical Universe Jim Blinn produced the first series of animation. Jim Blinn has also published the technique of bump mapping.
- **1979:** Ray tracing created at Bell Laboratory & Cornell University.
- **1980:** The first digital computer was used in computer graphics in the Digital Equipment Corporation (DEC).
- **1981:** The making of computer graphics for IMAX film format is done by Nelson Max at Lawrence Liver more National Laboratory. The Donkey Kong video game was introduced by Nintendo.
- **1982:** The first broad use of 3D graphics animation was done in Disney featured film.
- **AutoCAD 1.0** is launched-It is only used for wire frame representation.
- **1985:** Medical imaging software combined with Voxel technology.

History

- **1987:** Video graphics array (VGA) standard was introduced.
- **1989:** Super video graphics array (SVGA) was recommended. Tim Berners Lee developed the first website ever, which has the original URL (Universal resource locator).
- **1993:** Mosaic, the web browser was released by UIUC for general usage. The Codename of mosaic was "Mozilla."
- The **Mosaic**, the first web browser was released. First public call made by cell phone.
- **1994:** Netscape founded by developers of the Mosaic.
- **1995:** First, fully CGI (Computer-generated imagery) was released. MS Internet Explorer 1.0 released.
- **2000:** The first web-based CAD system Sketchup released.
- **2006:** Google acquires Sketchup.
- **2009:** The state of the art of computer graphics, as of 2009, was brief in a short video.
- **2013:** Now, it is possible to create graphics on a home computer.
- **2015:** **Big data** is being used to create animations.
- **2018:** Now, we can create "realistic" graphics on mobile phones. We can also create a completely CGI-based human face in real-time.

3D Graphics Techniques and Terminology

- **Modeling** :- Representing 3D objects (polygons, quads, etc.)
- **Rendering** :- Constructing 2D images from 3D models
- **Morphing** :- shifting pixels from one location to another
- **Rasterization**:- geometry to pixels
- **Blending** :- used to combine the color of a source pixel with the color of a destination pixel to produce a new color.

Cont..

- **Transformation** :- (translation, rotation, scaling, shearing)
- **Animation** :- simulating changes over time
- **Shading**:- modeling light interaction with 3D objects.
- **Hidden surface removal**:-display only visible part of an object.
- **Vertex**:- Point in 3D
- **Edge**:-line in 3D connecting two vertices

Cont..

- **Polygon/Face/Facet**:-arbitrary shape formed by connected vertices, **fundamental unit** of 3D computer graphics
- **Mesh** :-set of connected polygons forming a surface (or object)
- **Aliasing**: distortion object produced when representing a high-resolution signal at a lower resolution.
- **Anti-aliasing**:- a technique used to remove aliasing.
- **Texture mapping**:- covering a geometry using images. A texture is simply a picture that we map to the surface of a polygon

Example of Aliasing and Anti-aliasing



**Aliased polygons
(jagged edges)**



Anti-aliased polygons

Advantage of computer graphics

- Computer graphics is one of the most effective and commonly used ways of communication with computer.
- It provides tools for producing **pictures of “real-world” as well as synthetic objects** such as mathematical surface in 3D
- It has ability to show moving pictures thus possible to produce animations with computer graphics.

Cont..

- With the use of computer graphics, we can control the animation by adjusting the speed, portion of picture in view the amount of detail shown and so on.
- It provides tools called motion dynamics.
 - In which user can move objects as well as observes as per requirement.
 - For example walk throw made by builder to show flat interior and surrounding.

Major Applications Area

- Almost any attempt can make some use of computer graphics, but the major consumers of computer graphics technology include the following industries:
 - ❑ **Video games**- increasingly use sophisticated 3D models and rendering algorithms.
 - ❑ **Cartography** - Drawing maps
 - ❑ **Typography** - the use of character images in publishing - replacing hard type of the past.

Cont..

- **Cartoons** -are often rendered directly from 3D models
 - Many traditional 2D cartoons use **backgrounds rendered** from 3D models which allows a continuously moving viewpoint without huge amounts of artist time.
- **Weather Maps** – Real-time mapping, symbolic representations.
- **Satellite Imaging** – Geodesic images
- **Entertainment** – Movies and games.
- **Business presentation graphics** – A picture is worth a thousand words".

Cont..

- ❑ **Film special effects** use almost all types of computer graphics technology.
 - Almost every modern film uses digital compositing to superimpose backgrounds with separately filmed foregrounds.
 - Many films use computer-generated foregrounds with 3D models
- ❑ **Photo Enhancement** - Sharpening blurred photos.
- ❑ **Animation**- is a technique to create an illusion of motion through sequences of images.

Cont..

- **CAD/CAM:** CAD/CAM (Computer-aided design and computer-aided manufacturing).
 - Used to design and build prototypes, finished products, and manufacturing processes.
- **Simulation** - can be thought of as accurate video gaming.
 - For example, **a flight simulator uses sophisticated 3D graphics** to simulate the experience of flying an airplane.
 - Such simulations can be extremely useful for initial training in **safety-critical domains**
 - such as driving, and for scenario training for experienced users such as **specific fire-fighting situations** that are too costly or dangerous to create physically.

Cont..

- **Medical Imaging:** MRIs, CT scans, and other internal scans are possible because of computer graphics
 - Computer graphics is used to create shaded images that help doctors digest the most salient information from such data.
- **Presentations:** Computer graphics are used for making charts, bar diagrams, and other images for the presentation purpose, with the graphical presentation the user, can easily understand the points

cont..

- **Engineering Drawing** - mechanical, electrical, civil, etc. - replacing the blue prints of the past
 - **Architecture** - Construction plans, exterior sketches - replacing the blue prints and hand drawings of the past.
- **Art** - Computers provide new medium for artists.
- **Training** - Flight simulators, Medical training , computer aided instruction, etc.
- **Computer graphics user interfaces GUIs** - A graphic, mouse-oriented paradigm which allows the user to interact with a computer.

Cont..

- **Information visualization**- creates images of data that do not necessarily have a "natural" visual depiction.
 - For example, the temporal trend of the price of ten different stocks does not have an obvious visual representation, but clever graphing techniques can help humans find patterns in such data.
- **Virtual Reality**- attempts to immerse the user into a 3D virtual world.
 - For true virtual reality, sound and force feedback should be provided as well.

END

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Chapter Two

Graphics Hardware

Video Display Devices

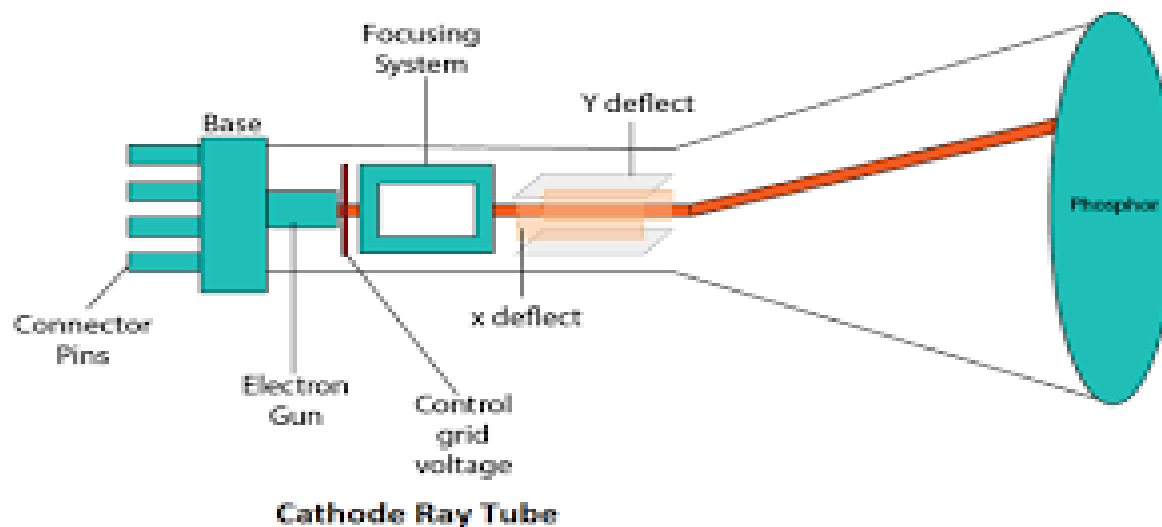
- Graphics hardware is the hardware components to quickly render 3D objects as pixels on your computer screen.
- Display devices are also known as output devices.
- Most commonly used output device in a graphics system is a video monitor.
 - They work to produce images on screen.
- Graphics Hardware includes
 - Display (CRT, LCD, LED, Plasma panels ...)
 - Graphics accelerator
 - Scan controller
 - Video Memory (frame buffer)
 - Display/Graphics Processor
 - CPU/Memory/Disk ...
 - Today's graphics hardware do have great processing and rendering capacity.

Cathode Ray Tube monitor CRT

- The primary output device in a graphical system is the video monitor.
- The main element of a video monitor is the Cathode Ray Tube (CRT), shown in the following illustration.
- The operation of CRT is very simple

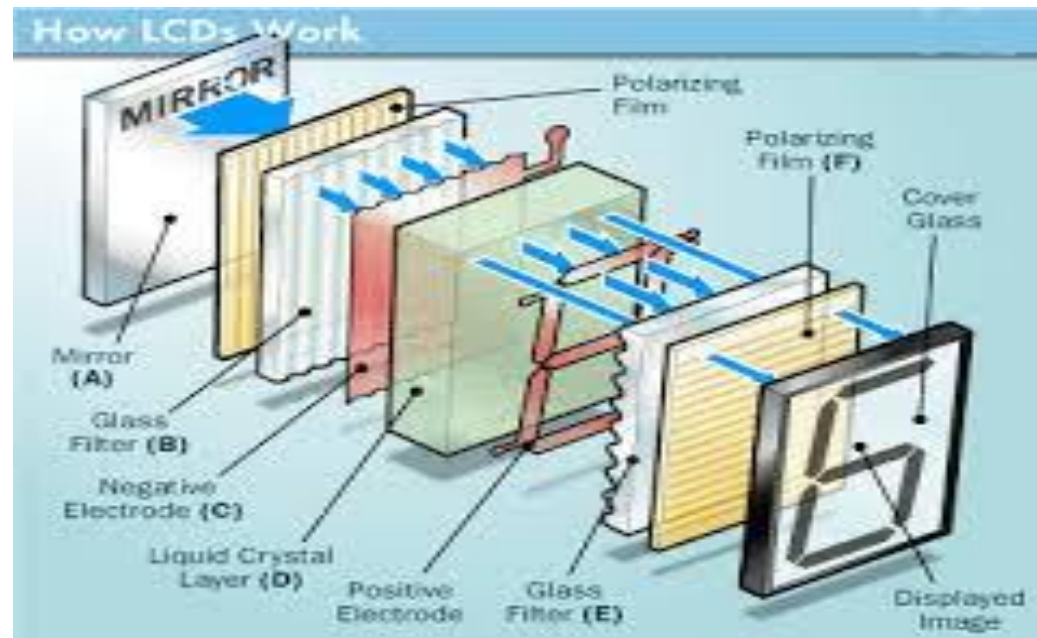
Operation of CRT

- The electron gun emits a beam of electrons (cathode rays)
- The electron beam passes through focusing and deflection systems that direct it towards specified positions on the phosphor-coated screen.
- When the beam hits the screen, the phosphor emits a small spot of light at each position contacted by the electron beam
- It redraws the picture by directing the electron beam back over the same screen points quickly.



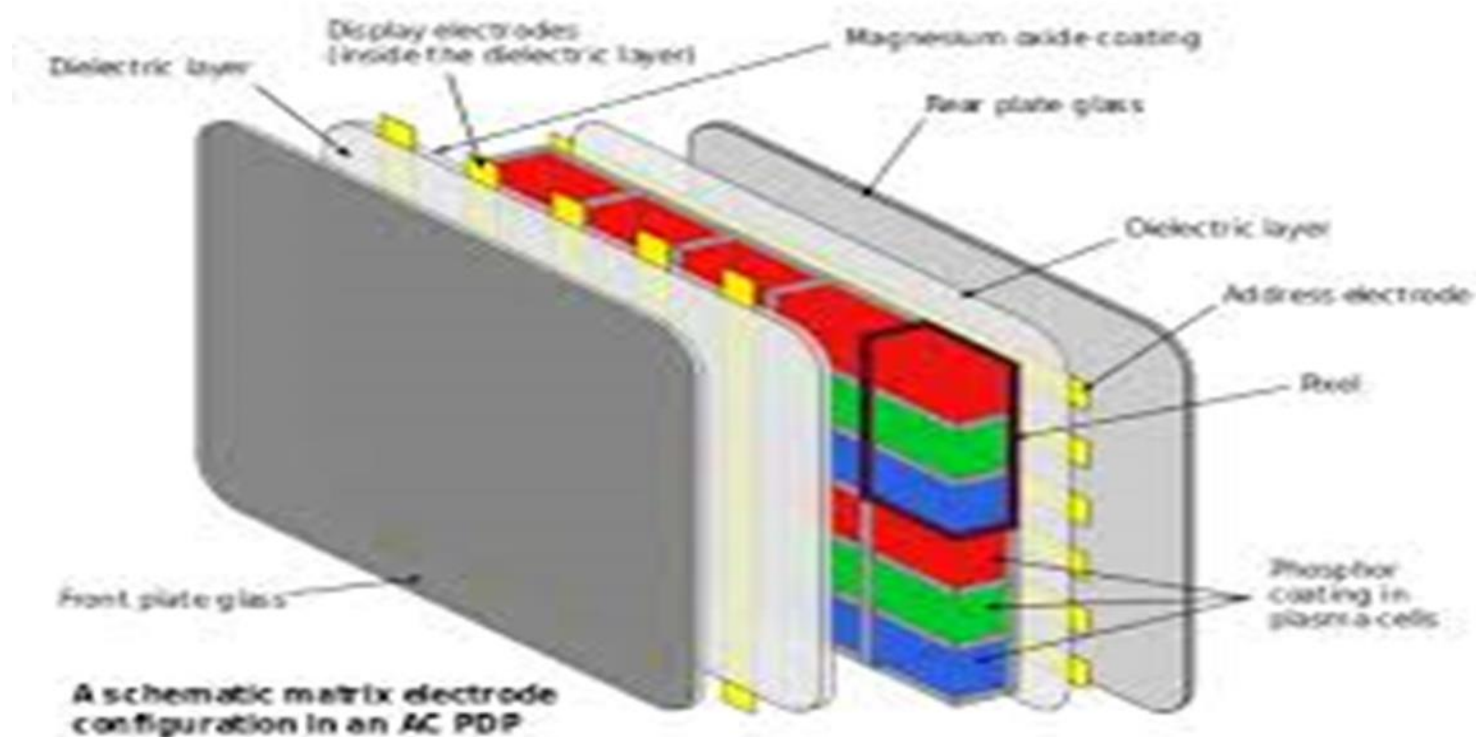
Liquid Crystal Display (LCD)

- Liquid crystal refers to the fact that these compounds have a crystalline arrangement of molecules, yet they flow like a liquid.
- Flat panel displays use nematic liquid crystal



Plasma Panels Display

- Picture definition is stored in a refresh buffer, and the firing voltages are applied to refresh the pixel positions 60 times per second.



Properties of video monitor

- Pixels
- Resolution
- Aspect ratio
- Frame Buffer
- Bitmap Vs Pixmap
- Frame
- Refresh Rate

Pixels

- In all the monitors mentioned above, the display area (screen) is considered as a series of picture elements called **pixels** (picture element or pel) arranged in rows and columns.
 - A pixel is made up of a **RED**, **GREEN** and **BLUE** components (RGB).
- Each component glows in different intensities based on their intensity values retrieved from the frame buffer.
 - The color of the pixel is the **combined** colors of all the **three** component intensities.
- As each pixel is smaller than the tip of a pin our eyes cannot distinguish the component colors but it gets the combined **effect** of all the three colors

Resolution

- Is the maximum number of points that can be displayed without overlap on a CRT
- Typical resolution on high-quality systems is 1280 x 1024 (i.e., 1280 columns by 1024 rows) with higher resolutions available on many systems.
- High resolution systems are often referred to as **high-definition systems**.
- The physical size of a graphics monitor is given as the length of the screen diagonal, with sizes varying from about 12 inches to 27 inches or more.

Aspect Ratio

- Another property of video monitor is aspect ratio.
- This number gives the ratio of vertical points to horizontal points necessary to produce equal-length lines in both directions on the screen.
 - (Sometimes aspect ratio is stated in terms of the ratio of horizontal to vertical points.)
- An aspect ratio of 3:4 means that a vertical line plotted with three points **has the same length as a horizontal** line plotted with **four points**.

Frame Buffer

- Picture definition is **stored in a memory** area called the refresh buffer or frame buffer.
- This memory area holds the set of intensity values for all the screen points (pixel).
 - For e.g., an intensity value for a single pixel requires 24 bits in a color display (i.e., 24 bits = 8 bits for **RED** +8 bits for **GREEN**+ 8 bits for **BLUE**).
- Stored intensity values are then retrieved from the refresh buffer to plot a pixel color.

Bitmap

- In a simple black-and-white system, each screen point is either on or off, so only one bit per pixel is needed to control the intensity of screen positions.
 - For e.g in a bi-level system, a bit value of 1 indicates a BLACK dot and a value of 0 indicates a WHITE dot. On a black-and-white system with one bit per pixel, the frame buffer is commonly called a **bitmap**.

Pixmap

- Additional bits are needed when color and intensity variations can be displayed.
 - 24 bits per pixel and higher are included in high-quality systems, which can require several megabytes of storage for the frame buffer, depending on the resolution of the system.
- A system with 24 bits per pixel and a screen resolution of 1024 by 1024 requires 3 megabytes (3MB) of storage for the frame buffer.
 - For systems with multiple bits per pixel, the frame buffer is often referred to as a **pixmap**

Frame

- A frame is a **single, still image** displayed by the computer.
- E.g. When the display monitor finishes plotting intensity starting from left top pixel to right bottom pixel, we say one FRAME is completed.

Refresh rate

- Refresh rate in computer graphics refers to the number of times per second that a display hardware updates its buffer.
- It is measured in Hertz (Hz).
 - For instance, a refresh rate of 60 Hz means the display updates 60 times per second.

Types of Image Production in Monitors

1. Raster-Scan Method

2. Random-Scan Method (Vector Scan method)

Raster-Scan Method (bitmap)

- The **most common type** of method used in graphics monitors
- The intensities are plotted across the screen **one row at a time from top to bottom** (called scan conversion or rasterization).
- The electron beam moves across each row, the beam intensity is turned on and off to create pattern illumination
- Each pixel in **a row from left to right is plotted with the intensity** value retrieved from the frame buffer. Each pixel row creates a pattern of illuminated spots.
- The picture definition is stored in a memory area called the *refresh buffer or frame buffer*.

Random-Scan Method (Vector Scan method)

- When operated as a random-scan display unit, a monitor plots pixel only to the parts of the screen where a picture is to be drawn.
- Random scan monitors draw a picture one line at a time and for this reason are also referred to as vector displays (*or stroke-writing or calligraphic displays*).
- The component lines of a picture can be drawn and refreshed by a random-scan system in any specified order.

Cont...

- A pen plotter operates in a similar way and is an example of a random-scan, hard-copy device.
- **Picture** definition is now **stored** as a set of line drawing commands in an area of memory referred to as the refresh display file.
 - Sometimes the refresh display file is called the *display list*, *display program*, or simply the *refresh buffer*.

Figure-1 : Raster scan method
used in a CRT monitor

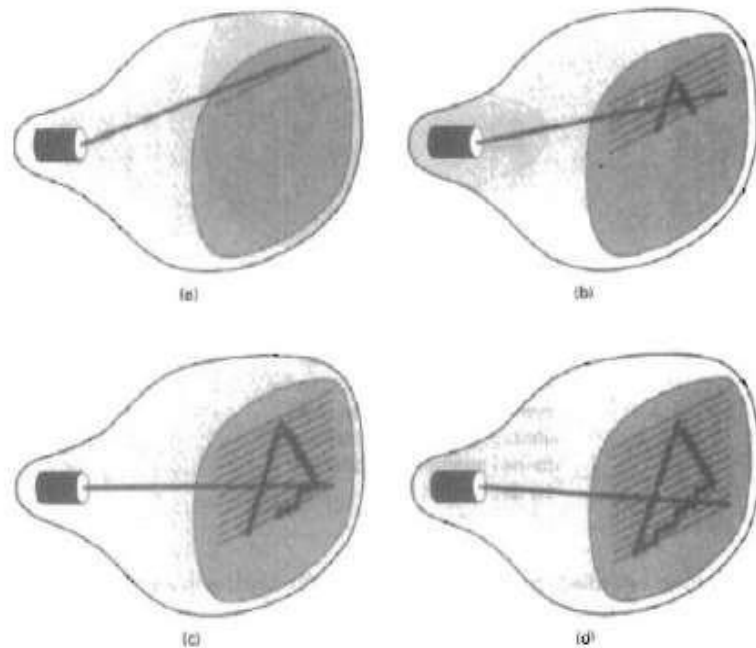
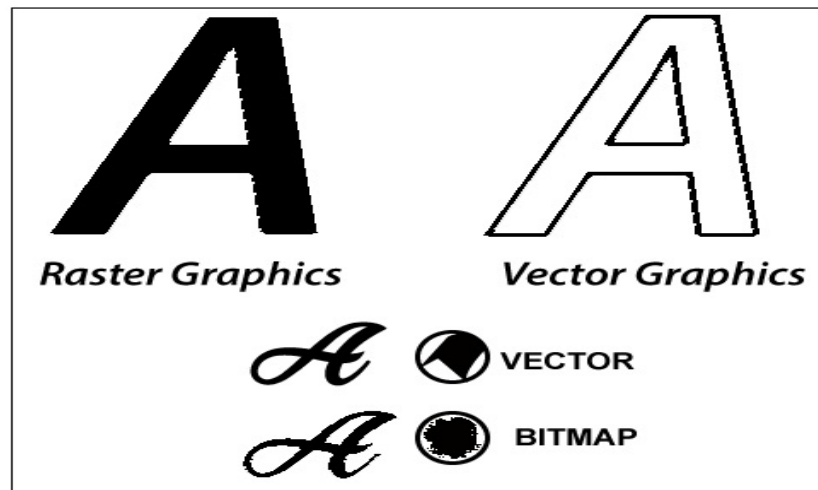
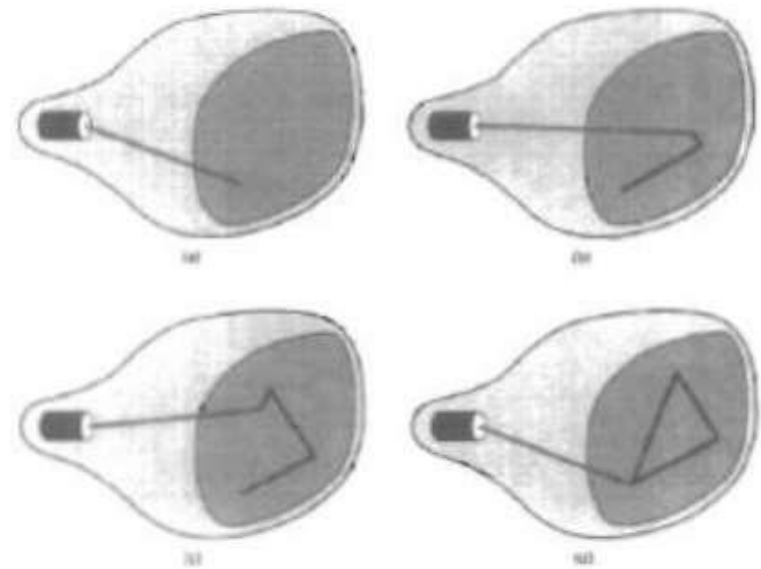


Fig 2 Vector Scan method



Introduction to the 3D graphics pipeline

- The 3D graphics pipeline is a sequence of steps used to create a 2D image from a 3D scene.
- The graphics pipeline typically accepts some representation of a 3D scene as an **input** and **results** in a 2D raster image as output.
- This process involves several stages, each transforming the scene data to progressively more advanced representations until a final image is rendered.
- Understanding this pipeline is essential for developing and optimizing 3D graphics applications, such as video games, simulations, and virtual reality.

Stages of the graphics pipeline:

- Modeling transformation
- Lighting
- Viewing transformation
- Clipping
- Scan conversion or Rasterization
- Texturing or shading
- Display

Modeling transformation

- In this stage the 3D geometry provided as input is **established**
- Applying transformations such as scaling, rotation, and translation to position objects in the scene.

Lighting

- Geometry in the complete 3D scene is lit according to the defined locations of light sources and reflectance and other surface properties.
- Per-pixel lighting can be done on modern graphics hardware as a post-rasterization process by means of a fragment shader program.

Viewing transformation

- Objects are transformed from 3D world-space coordinates into a 3D coordinate system based on the position and orientation of a virtual camera.
- This results in the original 3D scene as seen from the camera's point of view, defined in what is called eye-space or camera-space.

Projection transformation

- Projection transformation is a crucial step in the graphics pipeline that converts 3D coordinates into 2D coordinates, allowing for the creation of a view that can be displayed on a 2D screen.

Clipping

- Clipping is not necessary to achieve a correct image output, but it accelerates the rendering process by eliminating the unneeded rasterization and post-processing on primitives that will not appear anyway.

Scan conversion or rasterization

- Rasterization is the process by which the 2D image-space representation of the scene is converted into raster format and the correct resulting pixel values are determined.

- **Texturing or shading**

- At this stage of the pipeline individual fragments (or pre-pixels) are assigned a color based on values interpolated from the vertices
- To add detail, color, and realism to 3D objects by applying images (textures) to their surfaces.

- **Display**

- The final-colored pixels can then be displayed on a computer monitor or other display.

Graphics software and standard

- There are mainly two types of graphics software:
 - General programming package
 - Special-purpose application package

General programming package

- A general programming package provides an **extensive set of graphics function** that can be used in high level programming language
 - such as **C, C++ or FORTRAN**.
- It includes basic drawing element shape like line, curves, polygon, color of element transformation etc.
 - Example: - **GL (Graphics Library)**.

Special-purpose application package

- Special-purpose application package are customize for particular application
 - Which implement required facility and provides interface so, that user need not to worry about how it will work (programming).
 - User can simply use it by interfacing with application.
- Example: - **CAD**, medical and business systems.

Graphics Function

- A general purpose graphics package provides user with Variety of function for creating and manipulating pictures.
- The basic building blocks for pictures are referred to as output primitives.
 - They includes **character**, **string**, and **geometry entities** such as **point**, **straight lines**, **curved lines**, **filled areas** and **shapes** defined with arrays of color points.

Cont...

- Input functions are used for control & process the various input device such **as mouse, tablet, etc.**
- Control operations are used to controlling and **housekeeping tasks** such as clearing **display screen** etc.

NB All such inbuilt function which we can use for our purpose are known as **graphics function**

Software Standard

- Primary goal of standardize graphics software is portability
 - so that it can be used in any hardware systems & avoid rewriting of software program for different system
- **Graphical Kernel System (GKS)**
 - It is the first graphics software standard by the international standard organization (ISO) and various national standard organizations including ANSI
 - GKS was originally designed as the **two dimensional** graphics package and then later extension was developed for three dimensions.

PHIGS (Programmer's Hierarchical Interactive Graphic Standard)

- PHIGS is extension of GKS.
 - Increased capability for object modeling, color specifications, surface rendering, and picture manipulation are provided in PHIGS.
- Extension of PHIGS called "PHIGS+" was developed to provide three dimensional surface shading capabilities not available in PHIGS.

End

Chapter Three

Introduction to rendering process with Open GL

Introduction to Open GL

- It is a software interface for **graphics hardware**
- OpenGL renders **multidimensional** objects into a frame buffer.
- OpenGL is industry-standard graphics software
 - Programmers can create **high-quality motionless and animated three-dimensional** colour images.
- OpenGL is the best environment for developing portable, interactive **2D and 3D** graphics applications.
 - Since its introduction in 1992, OpenGL has become the industries

Cont..

- OpenGL is a computer graphics rendering application programming interface, or API (for short)
 - Most widely used API
 - With it, you can generate high-quality color images by rendering with geometric and image primitives
 - It forms the basis of many interactive applications that include 3D graphics
 - By using OpenGL, the graphics part of your application can be
 - operating system independent
 - window system independent

OpenGL

- **Simplifies Software Development**
 - Open GL routines simplify the development of graphics software
- **Developer Audience:** Designed for use by C/C++ /java programmers
- **Available everywhere**
 - supported on all UNIX workstations, and shipped standard with every Windows 95/98/2000/NT/2007/20010/11 and MacOS PC
- **High Visual Quality and Performance**
 - Visual computing application requiring maximum performance.
 - These capabilities allow developers in diverse markets such as broadcasting , CAD/CAM/CAE, entertainment, medical imaging, and virtual reality

OpenGL Programming

- For writing a program, using OpenGL, for any of the operating systems we can use **OpenGL Utility Library** named, **"glut"**.
- **"glut"** is used to initialize OpenGL on any platform
 - e.g. Microsoft Windows or Linux etc because it is platform independent.
- **"glut"** can create window, get keyboard input and run an event handler or message loop in our graphics application.
- All those functions that start with the prefix **"gl"** are the core OpenGL functions and those which start with **"glu"** or **"glut"** are the OpenGL utility library functions.
- We will See ALL OpenGL Functions in **LAB Section**

Coordinate Systems

- When you draw **lines** or **plot** points on a simple flat computer screen, you specify a position in terms of a row and column.
 - For example, a standard VGA screen has **640** pixels from left to right and **480** pixels from top to bottom.
 - To specify a point in the middle of the screen, you specify that a point should be plotted at (320,240)—
 - That is, **320** pixels from the left of the screen and **240** pixels down from the top of the screen.
- To specify a point in the middle of the screen, you specify that a point should be plotted at (320,240)—that is, **320** pixels from the left of the screen and **240** pixels down from the top of the screen.
 - Let us see 2D

2D Cartesian Coordinates

- Cartesian coordinates are specified by an x coordinate and a y coordinate
 - The x coordinate is a measure of position in the **horizontal** direction, and y is a measure of position in the vertical direction.
- The *origin* of the Cartesian system is at $x=0, y=0$
- The x -axis and y -axis are perpendicular (intersecting at a right angle) and together define the xy plane.
 - A *plane* is, most simply put, a flat surface
 - In a system with only two axes, there is naturally only one plane to draw on.

2D The Cartesian plan

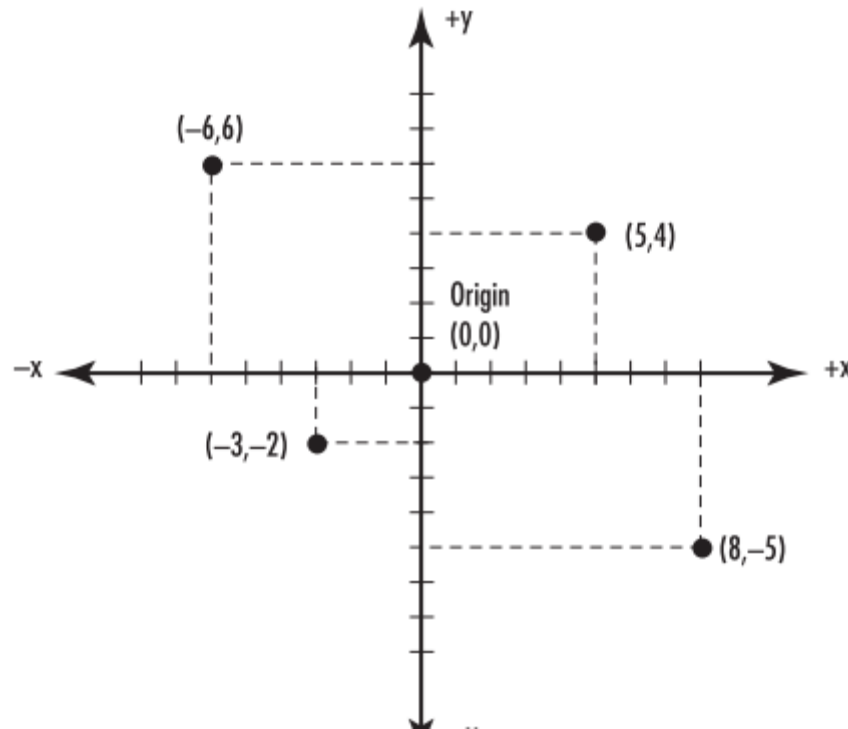


FIGURE The Cartesian plane

Coordinate Clipping

- A window is measured physically in terms of pixels.
- Before you can start plotting points, lines, and shapes in a window, you must tell OpenGL how to translate specified coordinate pairs into screen coordinates.
- You do this by specifying the region of Cartesian space that occupies the window; this region is known as the clipping region
- In two-dimensional space, the clipping region is the minimum and maximum x and y values that are inside the window. Example

Cont

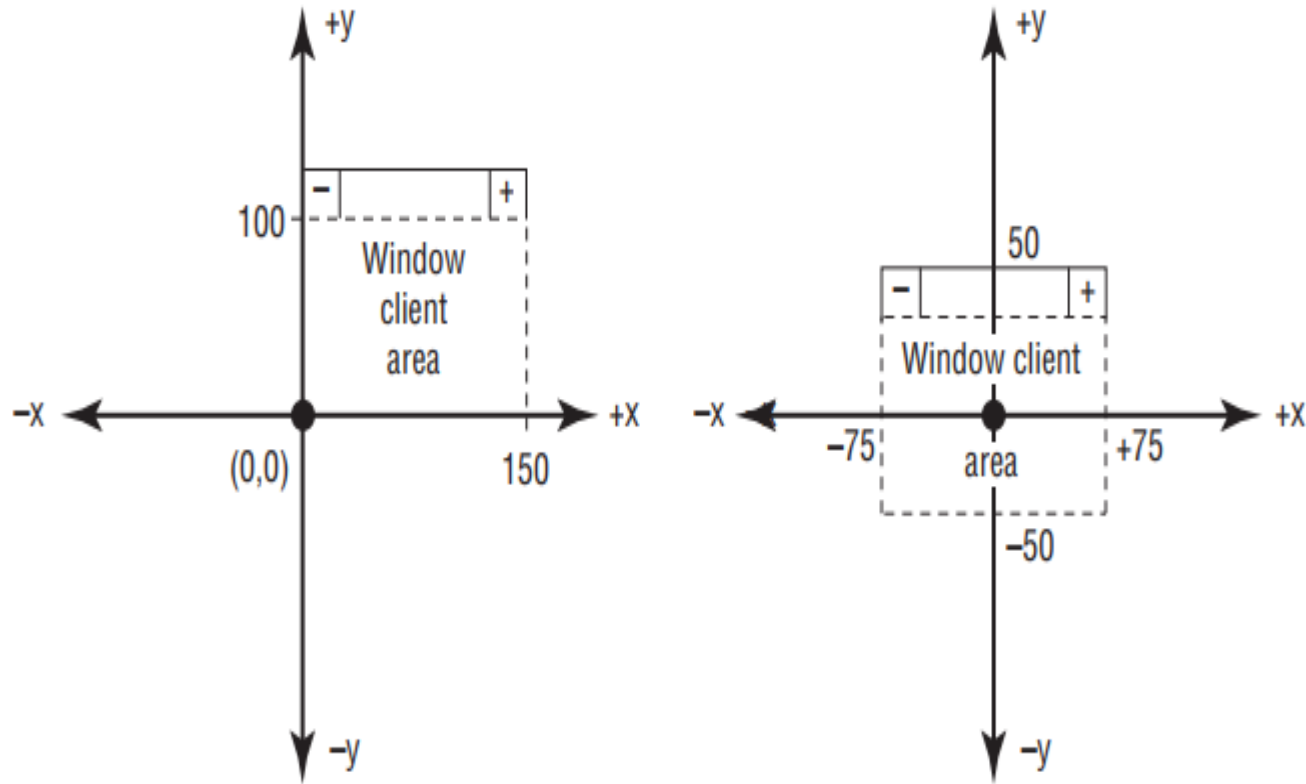


Figure Two clipping regions

- The primary use of clipping in computer graphics is to remove objects, lines, or line segments that are outside the viewing pane.

Viewports

- Rarely will your clipping area width and height exactly match the width and height of the window in pixels
- The coordinate system must therefore be mapped from **logical Cartesian coordinates** to physical screen pixel coordinates.
- This mapping is specified by a setting known as the **viewport**.
 - *Where the area picture to be displayed*
- The viewport is the region within the window's client area that is used for drawing the clipping area
- The viewport simply maps the clipping area to a region of the window.
- Usually, the **viewport** is defined as the entire window, but this is not strictly necessary; for instance, you might want to draw only in the lower half of the window

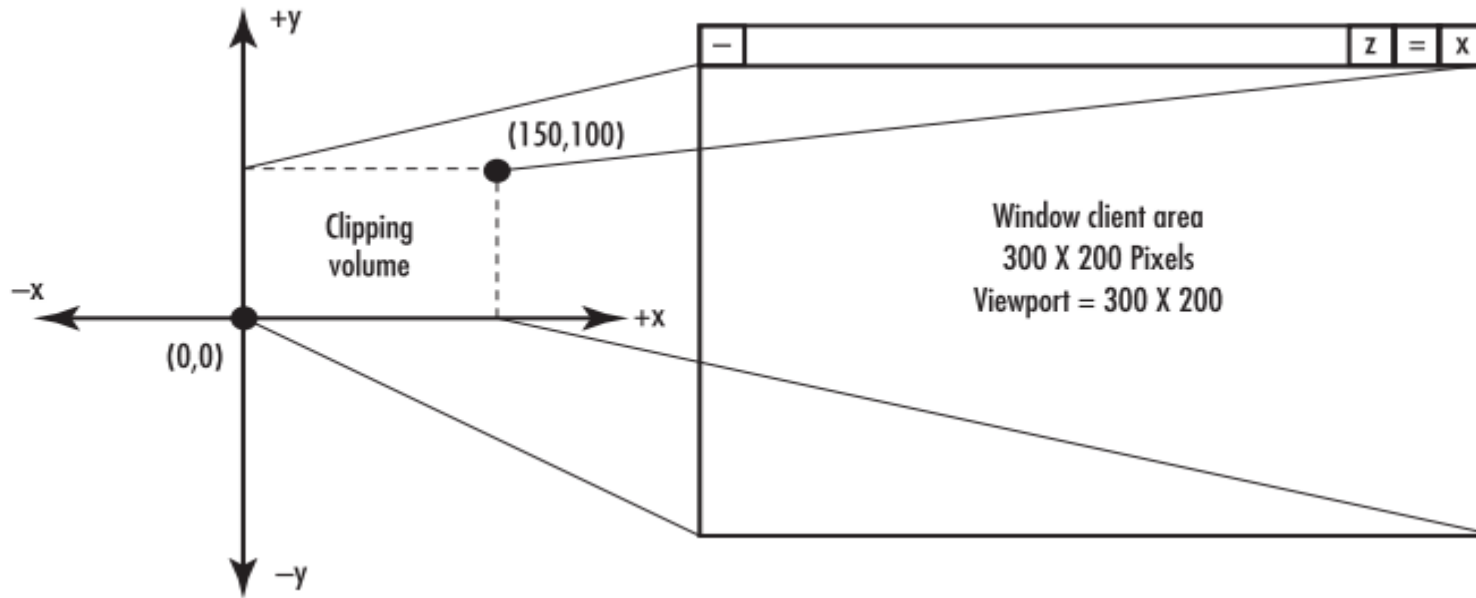


Figure-1 viewport defined as twice the size of the clipping area.

- A Window measuring 300×200 pixels with the viewport defined as the entire client area.
- Clipping area set to 0 to 150 along the x-axis and 0 to 100 along the y-axis, the logical coordinates would be mapped to a larger screen coordinate system in the viewing window.
- Each increment in the logical coordinate system would be matched by two increments in the physical coordinate system (pixels) of the window

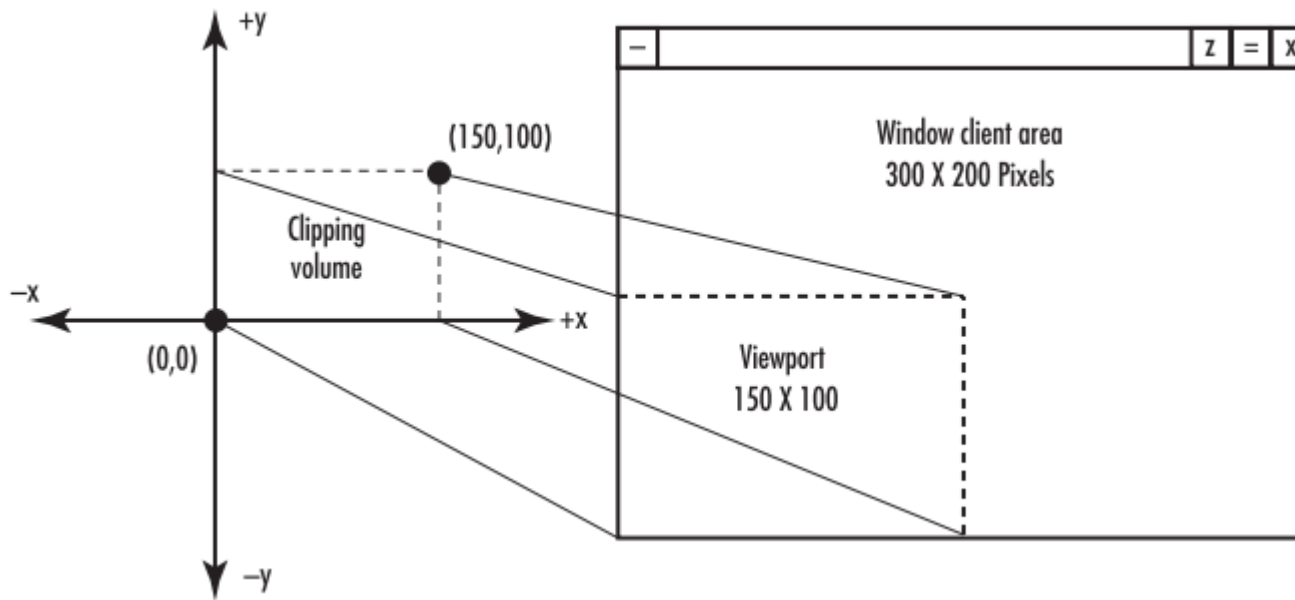
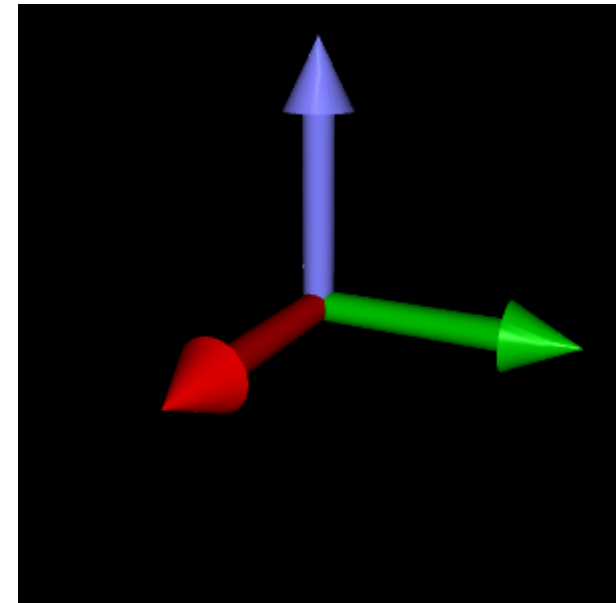
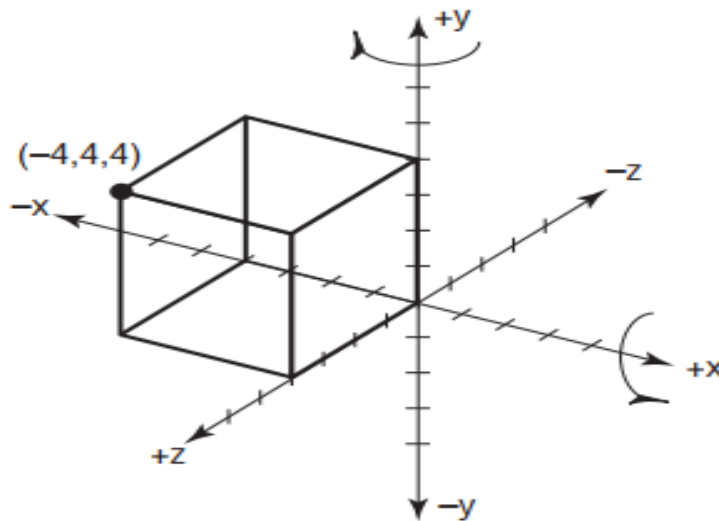


Fig 2 A viewport defined as the same dimensions as the clipping area

- The viewing window is still 300×200 pixels, however, and this causes the viewing area to occupy the lower-left side of the window.
- You can use viewports to **shrink** or **enlarge** the image inside the window and to display only a portion of the clipping area by setting the viewport to be larger than the window's client area.

3D Cartesian Coordinates

- The Cartesian coordinate system with a new axis, z
- The z -axis is perpendicular to both the x - and y -axes
- It represents a line drawn perpendicularly from the center of the screen heading toward the viewer



Figure

Cartesian coordinates in three dimensions.

Output Primitives and Attributes

- In graphics, output primitives and attributes are fundamental concepts used to represent and manipulate graphical objects on a display

Output Primitives

- Output primitives are the basic elements used to create more complex graphical images
- **Points:** Represent a single location in a coordinate system.
- **Lines:** Defined by two endpoints and represent the shortest distance between them
- **Polylines:** A series of connected line segments
- **Circles:** Defined by a center point and a radius
- **Ellipses:** Defined by a center point and two radii (major and minor)

Primitives cont..

- **Arcs:** A portion of a circle or an ellipse.
- **Rectangles:** Defined by opposite corners or a center point and dimensions.
- **Polygons:** Closed shapes defined by multiple points connected by line segments.
- **Text:** Characters or strings that can be displayed in a graphical interface.
- **Images:** Bitmap or raster graphics, composed of pixels.

Attributes

- Attributes define the appearance and behavior of output primitives.
- **Color:** Specifies the color of a primitive. This can be in RGB, CMYK, or other color models.
- **Line Style:** Defines the appearance of a line (solid, dashed, dotted, etc.).
- **Line Width:** Specifies the thickness of a line.
- **Fill Style:** Defines how the inside of a shape is filled (solid color, pattern, gradient, etc.).
- **Fill Color:** Specifies the color used to fill shapes.
- **Text Font:** Defines the font family, size, style (bold, italic), etc., for text primitives.
- **Text Alignment:** Specifies how text is aligned relative to a reference point (left, center, right).
- **Transparency:** Defines the opacity level of a primitive.
- **Shading and Lighting:** Attributes that add depth and realism by simulating light interaction with surfaces.
- **Pattern and Texture:** Used to fill shapes with a specific pattern or texture rather than a solid color.

Examples

1. Line Drawing with Attributes:

1. Primitive: Line from (x_1, y_1) to (x_2, y_2)
2. Attributes: Color (red), Line Style (dashed), Line Width (2 pixels)

2. Filled Polygon with Attributes:

1. Primitive: Polygon with vertices at (x_1, y_1) , (x_2, y_2) , (x_3, y_3)
2. Attributes: Fill Color (blue), Fill Style (solid)

3. Text with Attributes:

1. Primitive: Text "Hello, World!" at (x, y)
2. Attributes: Font (Arial, 12pt, bold), Color (black), Alignment (center)

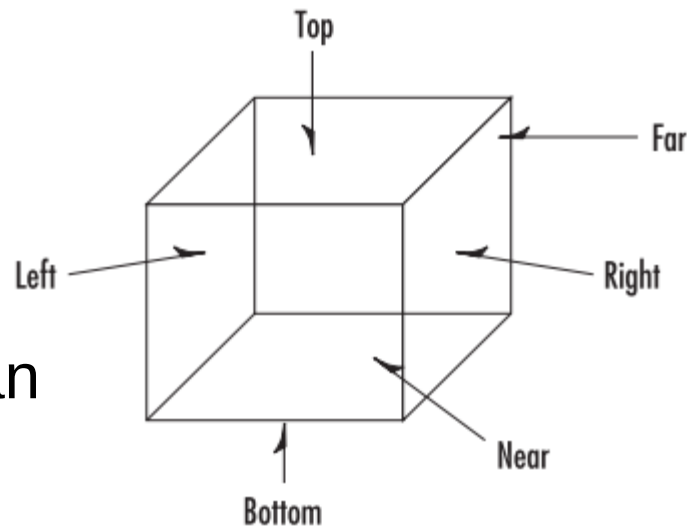
■ Projections

- Projection representing n dimensional object in $n-1$
- To display 3D objects on a 2D screen, Projection is required :
- Transforms 3D coordinates to 2D coordinates.
- Two Common types
 - Orthographic Projections
 - Perspective Projections

Orthographic Projections

- Orthographic projection is a method of showing three-dimensional objects in two dimensions (that is on plain surface).
- you are mostly concerned with two main types of projections in OpenGL.
 - The first is called an *orthographic*, or parallel projection.
- Typically, we use orthographic projections for 2D drawings, and we keep the z coordinate at 0.0 for our geometry

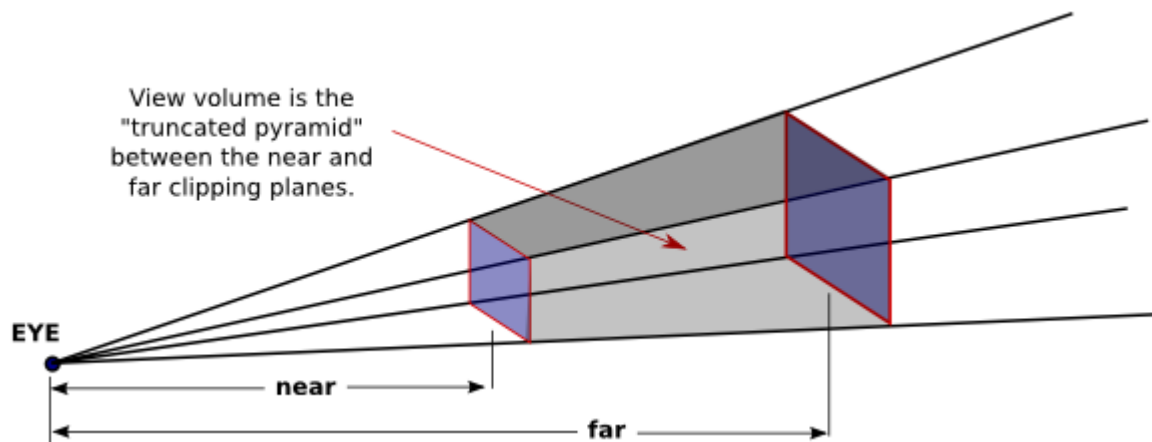
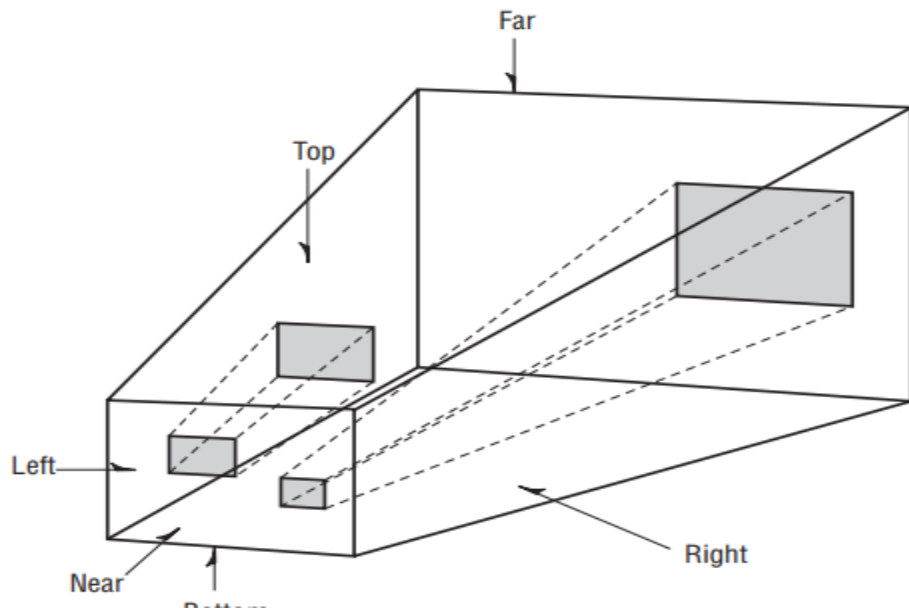
The clipping volume for an orthographic projection.

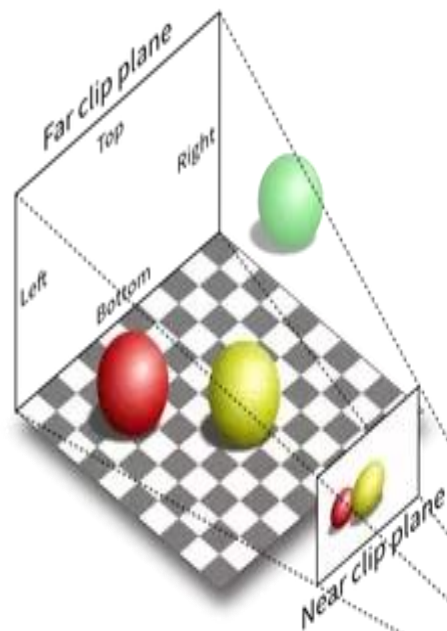


Perspective Projections

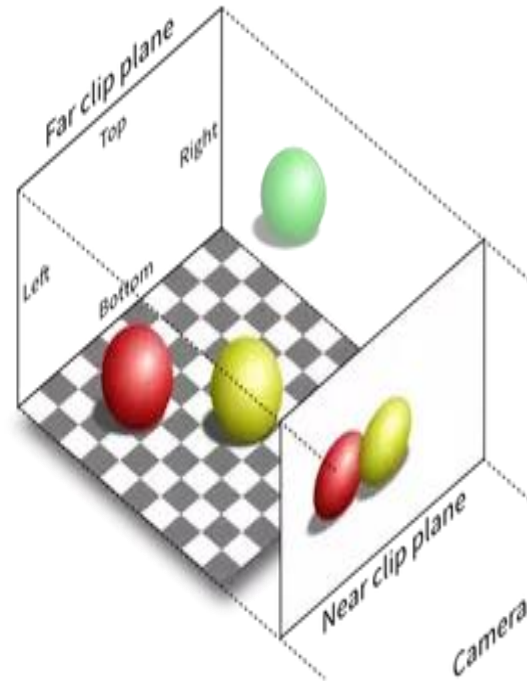
- The second and more common projection is the *perspective projection*.
- Perspective projection is a method of drawing three-dimensional objects generally on a flat surface, of an image as it is seen by the eye.
- This projection adds the effect that distant objects appear smaller than **nearby** objects.
 - The human eye perceives the objectives **nearer** to it bigger, and the objects farther, smaller
- This type of projection gives the most realism for simulation and 3D animation.
- In perspective projection **farther away object from the viewer, small it appears**. This property of projection gives an idea about depth.
- Perspective projection gives a more realistic representation of an object

Perspective





Perspective projection (P)



Orthographic projection (O)

THE END !!